

Instructor: Date:

Learner: Regional Branch:

Program: Center:

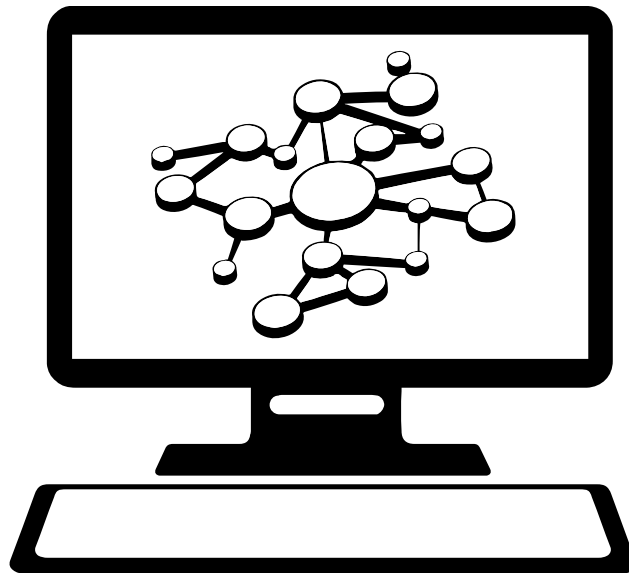
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Let's Start!

Using Simple Programming!



Learning Outcome:

Describe the development of software programming in simple spoken and written situations.

Ask your partner:

- What programming methods do you know?
- Do you have to use any specific programming language when using these methods? Why?

Can Do Statements:

- I can understand written information about the definition of pseudocode.
- I can understand written information about the importance of pseudocode.
- I can write pseudocode taking into consideration basic guidelines.





Let's Explore!

Today's presentation is about pseudocode and its importance in the system engineering field.



Instructions: Read the text below.

Jack: Thank you all for coming to our convention. Today's presentation is about pseudocode and its importance to our system engineering field. The information that I show you here is a case study which defines this concept as a tool to create programs in an easy way. Do not confuse this term with a programming language, it is more like a source of statements that helps us obtain the expected results in our program.

One effective way to solve these matters is using pseudocode, which is a tool to ensure that all the information required in a program is included. But, make no mistake, do not take the term "pseudocode" as a programming language, it is instead a simpler way to write code before using an actual language. In this way, this pre-coding method is really appealing for users since it offers a friendlier approach to achieve better programming results.

In this case, Professor Smith explains in his written document the importance of pseudocode. He says that programming can be a difficult task and that we usually use flowcharts to map out programs. However, flowcharts are not easy to modify, and they don't display all parts of a program. Pseudocode is important because it is more appealing for users since it has a friendly concept where you can write short phrases in English that later on can be translated into any language thanks to pseudocode. Don't complicate yourselves with difficult long statements, sometimes this makes things difficult and confusing.

To conclude the presentation, we need to understand the steps of a basic guideline to write pseudocode; so, later on, we start practicing. As I mention before, the first thing to do is to organize our ideas into short statements. In that way, it will be clear for us what we want to achieve. Don't forget using short phrases to write code for programs. The next step is to select some important words and capitalize them. So, we let the program know what we want to show. Do not capitalize everything; you need to identify the most relevant tasks for your program. Finally, we just need to indent the statements and make sure they are in shown in a loop.

Now that we review this information, it's time to practice.





Let's Practice!

The lecturer is defining what pseudocode is.

Instructions: Read the paragraph. Underline all the verbs in Simple Present tense.

Pseudocode makes reference to a programming tool that ensures that all the required information of a program is included. Some people confuse this term with a programming language, but it is instead a simpler way to write code before using an actual language.



Let's Practice!

Please, take these instructions into account!

Instructions: Read the sentences. Circle the correct option to complete each sentence.

1. (Do not forget / Do not make / Do not take) that the instructions used in pseudocode need to be universal.
2. (Do not read / Do not use / Do not think) any language specific-commands since this format is not used in compilation or execution.
3. (Do not forget / Do not say / Do not write) run on sentences. Each statement must go in independent lines.
4. (Do not capitalize / Do not standardize / Do not criticize) all the words in your statements, you just need to capitalize keywords such as Start, Read, End, etc.
5. (Do not take / Do not write / Do not forget) to indent your statements, this eases the reading of the flow of your program.



Extension Activity

Creating Pseudocode Samples

Instructions: Complete the following steps. Fill in the Self-Assessment.

- Look for more information about the steps for creating pseudocodes.
- Identify key ideas and examples.
- Think of a simple program that you would like to create.
- Create a pseudocode with the tasks that your program will perform.
- Share your pseudocode with a classmate.

Self-Assessment

Yes No

1. I identify information about the steps for creating pseudocodes.

☐ ☐

2. I recognize key ideas and examples.

☐ ☐

3. I create a pseudocode with the tasks that my program will perform.

☐ ☐

4. I share my pseudocode with a classmate.

☐ ☐


Target Words

- **pseudocode** (noun): simple code that can be translated into any programming language
e.g. One effective way to solve these matters is using pseudocode, which is a tool to ensure that all the information required in a program is included.
- **map out** (verb): to schematize something
e.g. Most of them usually use flowcharts to map out programs.
- **capitalize** (verb): to put the first letter of a word in upper case
e.g. Third, do not capitalize all the words in your statements, you just need to capitalize keywords such as Start, Read, End, etc.
- **indent** (verb): to provide an extra space at the beginning of a sentence
e.g. Finally, do not forget to indent your statements, this eases the reading of the flow of your program.
- **achieve** (verb): to reach established goals
e.g. In this way, this pre-coding method is really appealing for users since it offers a friendlier approach to achieve better programming results.



Learning Points

Negative Imperatives

You use **negative imperatives** to give instructions or commands about actions that you don't want other people to do.

Imperatives are useful to express others what we expect them to do or **not** to do. When we use imperative sentences **negative** in the negative form, we have to start the command with **do not** or **don't**. Structure:

Do not (Don't) + verb in base form + complement + period (.) or exclamation sign (!)

Here we have some examples:

Do not use long statements, make it simple!

Don't use long statements, make it simple.

At the end of the sentence, we need to add a period (.) or an exclamation sign (!); that depends on the communicative intention of the command.

Examples

- *But, make no mistake, do not take the term "pseudocode" as a programming language, it is instead a simpler way to write code before using an actual language.*
- *First, do not forget that the instructions used in pseudocode need to be universal. Do not use any language specific-commands since this format is not used in compilation or execution.*



Learning Points

Basic Guidelines to Use Pseudocode

Pseudocode is a simple method to write code statements when mapping out programs.

The main objective of using pseudocodes is to make programming easier for software developers since most of the times programs can be difficult and hard to elaborate.

The first step to using pseudocodes is to write what you need the program to show in English. This helps you organize your ideas. Make sure you use short statements. And do not include language-specific commands because pseudocodes must be universal. The second step is to divide the tasks that your program will perform by lines. The idea is not to overburden the processes with confusing information.

The next step is to make sure that important words are capitalized, so you will be able to recognize the main tasks in each instruction. Finally, make sure you indent statements in a loop, so you can see how the program flows.